Facial Emotion and Sleep Detection via Audio Feedback: An Assistive AI System

Meesa Rakesh,[a)] Jatin Chandra Gupta,[b)] CH Virinchi,[c)] Majji Jayesh,[d)] and Lekshmi C. R[e)]

Amrita School of Artificial Intelligence,
Coimbatore,
Amrita Vishwa Vidyapeetham,
India

Abstract. Facial Emotion Recognition (FER) has a key part to play in human-computer interaction, most notably in the creation of assistive technology. This paper offers an AI-powered system specifically for visually impaired persons that can determine facial emotions as well as states of sleep via real-time auditory feedback. A Convolutional Neural Network (CNN) based model is developed on an eight-class dataset involving a new "Sleep" class and reaches 60.81% test accuracy. Prominent performance was seen in major classes like "Happy" (F1-score: 0.81) and "Sleep" (F1-score: 0.91). Face detection is managed through Haar-cascade, whereas verbal feedback is provided through Google Text-to-Speech (gTTS), allowing users to hear emotional cues. Batch normalization, dropout, and data augmentation were used to enhance generalization and counter class imbalance. Real-time testing showed 60% precision in emotion detection. Future improvements will include adding transfer learning, more profound architectures, and multilinguality support to enhance performance and accessibility even more.

I. INTRODUCTION

Facial emotion recognition (FER) is a key technology in human-computer interaction, with applications spanning accessibility, healthcare, surveillance, entertainment, and automotive safety. By detecting human emotions through facial expressions, intelligent systems can enhance communication, personalize user experiences, and support individuals with physical or cognitive limitations. A particularly impactful use of FER lies in assistive technology for the visually impaired, where it can deliver real-time awareness of the emotional states of nearby individuals, bridging communication gaps and promoting social inclusion [1].

This paper introduces a novel for Facial Emotion and Sleep Recognition with Audio Feedback, capable of identifying eight emotional states: Angry, Disgust, Fear, Happy, Neutral, Sad, Sleep, and Surprise. The system utilizes a Convolutional Neural Network (CNN) trained on an augmented dataset incorporating a new "Sleep" class, thereby improving the system's ability to detect sleep. Real-time video input is processed using Haar cascade face detection, followed by facial emotion classification. For accessibility, the system delivers immediate audio feedback using Google Text-to-Speech (gTTS) and Pygame, vocalizing the detected emotion to the user. This feature is especially valuable for visually impaired individuals, enabling them to interpret social cues in real-time.

The system presented has several advantages, such as low-latency real-time processing, double functionality in one light-weight architecture, and real-world applicability in diverse domains. Its originality is derived from the combination of emotion recognition and sleep detection within one model, coupled with the utilization of audio narration to enhance accessibility. Applications range from assistive technology for the blind to smart monitoring and emotion-aware interfaces in educational systems, healthcare systems, and entertainment systems.

The remainder of this paper is organized as follows: Section II discusses related work on facial emotion recognition and assistive technologies. Section III outlines the proposed methodology, the data set preparation, the model architecture, and the audio feedback pipeline. Section IV reports experimental results and performance analysis and then the paper is concluded in Section V.

[a)]Electronic mail: cb.sc.u4aie24134@cb.students.amrita.edu
[b)]Electronic mail: cb.sc.u4aie24162@cb.students.amrita.edu
[c)]Electronic mail: cb.sc.u4aie24158@cb.students.amrita.edu
[d)]Electronic mail: cb.sc.u4aie24128@cb.students.amrita.edu
[e)]Corresponding author: cr_lekshmi@cb.amrita.edu

II. RELATED WORK

Facial emotion recognition (FER) through image classification has gained significant traction due to its applications in human-computer interaction, mental health monitoring, and intelligent surveillance [2]. Various approaches, ranging from classical machine learning models to sophisticated deep learning architectures, have been utilized to improve both accuracy and resilience.

Srivastav *et al.* (2022) demonstrated the use of OpenCV for facial emotion detection, showcasing the viability of classical image processing methods [3]. Rana *et al.* (2023) compared multiple face recognition techniques for emotion analysis, assessing their accuracy and performance [4]. Nautiyal *et al.* (2023) developed a real-time emotion recognition system using deep learning, focusing on performance in dynamic environments [5], while Zim (2023) presented a CNN-based emotion analysis pipeline using Python and OpenCV [6].

In healthcare, Hunt and Kim (2024) explored image and video analysis for autism screening, highlighting the potential of automated FER in clinical settings [7]. Joseph *et al.* (2024) introduced an innovative deep learning framework with Botox feature selection, enhancing emotion classification accuracy [8]. Avabradha *et al.* (2024) proposed a multimodal emotion detection system that combines speech and facial data through decision-level fusion [9]. More recently, Vignesh *et al.* (2025) explored audio-video integration for improved speech emotion recognition [10], while Sharma *et al.* (2025) proposed an end-to-end attention-based network for autonomous facial emotion detection [11].

Based on these attempts, our system makes use of a CNN-based FER model in TensorFlow and OpenCV. The system incorporates data augmentation, dropout, and batch normalization for better generalization. Moreover, the addition of a new "Sleep" class and real-time audio feedback improves the system's accessibility, especially for visually impaired individuals.

III. METHODOLOGY

This section presents the methodology adopted for the proposed Facial Emotion and Sleep Recognition with an Audio Feedback system. The approach integrates dataset preparation, model architecture, training strategies, and real-time audio feedback generation. A Convolutional Neural Network (CNN) serves as the backbone of the facial emotion classifier, while Google Text-to-Speech (gTTS) and Pygame handle the speech synthesis component.

Dataset Refinement and Extension

The baseline dataset is derived from FER-2013, [12] which comprises 48×48 grayscale facial images representing seven basic emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. It contains 28,709 training and 3,589 test samples. To improve functionality and address class imbalance, two key augmentations were made:

The first enhancement involved introducing a new **Sleep** class, consisting of 4,388 manually curated images from closed-eye and drowsy facial expressions. These images were obtained through web scraping and validated to ensure they exhibit characteristics of sleep such as closed or drooping eyes. Augmentation was applied using brightness/contrast adjustments (±20%) and random cropping (±5%) to simulate variations.

The second improvement involved augmenting the **Disgust** class, which originally had only 547 samples. It was expanded to 2,111 images using geometric transformations, including rotations (±15°) and horizontal flips, supplemented with additional web-sourced images. All images were preprocessed via grayscale normalization, histogram equalization, and resizing using bilinear interpolation to ensure a uniform input size of 48×48 pixels.

Class	Testing	Validation	Training
Angry	496	495	3962
Disgust	212	211	1688
Fear	513	512	4096
Happy	900	898	7191
Neutral	621	619	4958
Sad	609	607	4861
Sleep	440	438	3510
Surprise	401	400	3201

FIGURE 1. Distribution of samples across emotion classes, including the added Sleep class.

Convolutional Neural Network Architecture

Convolutional Neural Networks (CNNs) are deep learning models that are very efficient at image classification because of their ability to learn automatic hierarchical spatial feature extraction from unprocessed input data. The architecture used in this paper consists of sequential convolutional layers, pooling layers, and fully connected layers, which collectively allow the model to recognize and classify facial expressions.

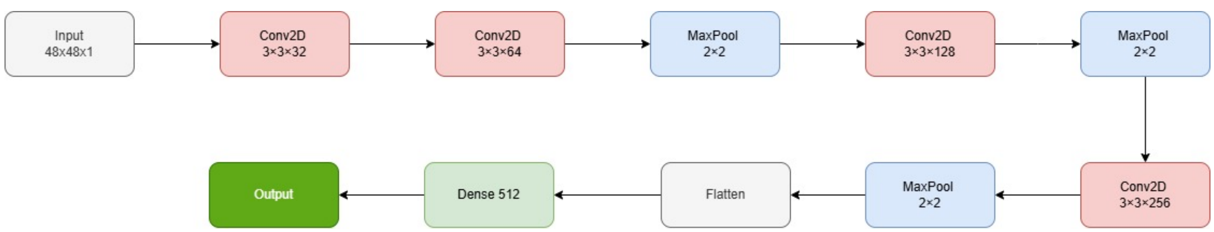

FIGURE 2. CNN architecture showing convolutional, pooling, and fully connected layers.

As illustrated in Figure 2, the convolutional layers apply a set of learnable filters across the input image. These filters scan the image spatially to generate feature maps, highlighting localized patterns such as edges, contours, and textures that are important for emotion recognition. The convolution operation is defined as:

$$F(x,y) = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} I(x+i, y+j) \cdot K(i,j) \tag{1}$$

In Equation 1, $I(x,y)$ denotes the input image, $K(i,j)$ is the convolution kernel (or filter) of size $m \times n$, and $F(x,y)$ represents the resulting feature map.

After every convolution step, a non-linear activation function is introduced to help the model capture complex patterns. This work uses the Rectified Linear Unit (ReLU), as defined in Equation 2.

$$f(x) = \max(0,x) \tag{2}$$

To downsample the feature maps, pooling layers like max pooling are employed. This process lowers computational load and enhances translation invariance by preserving the most significant features within localized areas.

```
Found 33467 images belonging to 8 classes.
Found 4180 images belonging to 8 classes.
Classes: {'angry': 0, 'disgust': 1, 'fear': 2, 'happy': 3, 'neutral': 4, 'sad': 5, 'sleep': 6, 'surprise': 7}
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_5 (Conv2D)	(None, 46, 46, 32)	320
batch_normalization_6 (BatchNormalization)	(None, 46, 46, 32)	128
conv2d_6 (Conv2D)	(None, 44, 44, 64)	18,496
batch_normalization_7 (BatchNormalization)	(None, 44, 44, 64)	256
max_pooling2d_3 (MaxPooling2D)	(None, 22, 22, 64)	0
dropout_4 (Dropout)	(None, 22, 22, 64)	0
conv2d_7 (Conv2D)	(None, 20, 20, 128)	73,856
batch_normalization_8 (BatchNormalization)	(None, 20, 20, 128)	512
max_pooling2d_4 (MaxPooling2D)	(None, 10, 10, 128)	0
dropout_5 (Dropout)	(None, 10, 10, 128)	0
conv2d_8 (Conv2D)	(None, 8, 8, 256)	295,168
batch_normalization_9 (BatchNormalization)	(None, 8, 8, 256)	1,024
max_pooling2d_5 (MaxPooling2D)	(None, 4, 4, 256)	0
dropout_6 (Dropout)	(None, 4, 4, 256)	0
flatten_1 (Flatten)	(None, 4096)	0
dense_2 (Dense)	(None, 512)	2,097,664
dropout_7 (Dropout)	(None, 512)	0
dense_3 (Dense)	(None, 8)	4,104

Total params: 2,491,528 (9.50 MB)
Trainable params: 2,490,568 (9.50 MB)
Non-trainable params: 960 (3.75 KB)

FIGURE 3. Model summary showing CNN layers and a total of approximately 2.49 million parameters.

After multiple rounds of convolution and pooling, the resulting feature maps are flattened and passed through fully connected layers. These layers act as a traditional neural network, learning complex relationships between high-level features and target classes. The final layer outputs class scores (logits), which are normalized using the Softmax function to generate probabilities for each emotion class [13]:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \tag{3}$$

Here, z_i is the logit for class i, C is the number of classes (in this case, eight), and $P(y_i)$ is the predicted probability that the input belongs to class i.

The training objective of the CNN is to minimize the categorical cross-entropy loss between predicted probabilities and true labels, as defined in Equation 4:

$$\mathscr{L} = -\sum_{i=1}^{C} y_i \log(P(y_i)) \tag{4}$$

Where y_i is the ground-truth label for class i, and $P(y_i)$ is the predicted probability as given by the Softmax function.

TABLE I. Classification Performance Metrics

Class	Precision	Recall	F1-Score	Support
Angry	0.43	0.48	0.45	496
Disgust	0.45	0.76	0.57	212
Fear	0.38	0.12	0.18	513
Happy	0.80	0.82	0.81	900
Neutral	0.46	0.61	0.52	621
Sad	0.45	0.32	0.37	609
Sleep	0.85	0.97	0.91	440
Surprise	0.67	0.72	0.69	401
Accuracy		0.60		4192
Macro Avg	0.56	0.60	0.56	4192
Weighted Avg	0.58	0.59	0.57	4192

Model Optimization and Training

To train the CNN model efficiently, categorical cross-entropy was used as the loss function, as it is well-suited for multi-class classification tasks by penalizing incorrect predictions based on the log difference between actual and predicted probabilities. The optimization process relied on the Adam (Adaptive Moment Estimation) optimizer, which adjusts learning rates individually for each parameter by maintaining moving averages of both the gradients and their squared values. This results in faster convergence and greater stability during training. To enhance learning efficiency and prevent stagnation, a learning rate scheduler (ReduceLROnPlateau) was applied, which reduces the learning rate when the validation loss stops improving. Additionally, training was made more robust using EarlyStopping and ModelCheckpoint callbacks—where the former halts training to avoid overfitting when no improvement is observed and the latter ensures the best-performing model is saved. Overall, the training strategy combined regularization, adaptive optimization, and monitoring to ensure high accuracy and generalization. The final architecture, shown in Figure 3, includes convolutional blocks with dropout and batch normalization, leading to a dense output layer, totaling approximately 2.49 million trainable parameters.

The final dataset was stratified into training (80%, 33,467 samples), validation (10%, 4,180 samples), and testing (10%, 4,192 samples) subsets, ensuring that the original class distributions were preserved across all splits. Each subset includes eight emotion classes: the seven standard categories—Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral—as well as the newly introduced Sleep category. Following data augmentation, the representation of the Disgust class increased from 0.8% to 5.05% of the total dataset, while the Sleep class accounts for approximately 6.3%. This rebalancing ensures more equitable representation across all categories, thereby enhancing model learning and reducing class bias during training.

Audio Feedback Generation

The audio feedback module operates alongside the CNN-based facial emotion and sleep recognition system to deliver real-time verbal outputs based on predicted emotional states. After a face is detected using OpenCV's Haar Cascade classifier, the region of interest is preprocessed and fed into the trained model for classification [14]. Upon predicting an emotion or sleep state, the corresponding label is converted into speech using the Google Text-to-Speech (gTTS) API. The resulting audio is played via the Pygame library. To ensure uninterrupted video streaming and prevent blocking during audio playback, speech synthesis and playback are executed in a separate thread. This multithreaded approach enables concurrent recognition and feedback, ensuring smooth and responsive interaction without compromising system performance.

IV. RESULTS AND DISCUSSION

The proposed system was evaluated across multiple dimensions, including classification accuracy, generalization ability, and performance in real-time scenarios. The Convolutional Neural Network (CNN) was trained on an augmented version of the FER-2023 dataset, enhanced with the addition of a new “Sleep” category. After completing 60 training epochs, the model demonstrated a stable convergence pattern with strong generalization properties.

The tableI shows the overall performance of our proposed model. The model demonstrates strong performance for the "Happy" and "Sleep" classes, with high precision, recall, and F1-scores (0.80/0.82/0.81 and 0.85/0.97/0.91, respectively), indicating reliable detection of these emotions. However, it struggles significantly with "Fear," showing poor recall (0.12) and a low F1-score (0.18), suggesting frequent misclassification. Classes like "Angry," "Neutral," and "Sad" show moderate performance, while "Disgust" has higher recall (0.76) but lower precision (0.45), implying many false positives. The overall accuracy of 0.59 and macro-average F1-score of 0.56 reveal that the model's performance is inconsistent across classes, likely due to imbalanced data (e.g., "Happy" has 900 samples vs. "Disgust" with 212). Improvements in handling minority classes and reducing false negatives for "Fear" could enhance model robustness.

The model achieved a final test accuracy of 60.81% and a test loss of 1.065. These results indicate the model's capacity to effectively recognize all facial emotions and also the introduced drowsiness-related "Sleep" category. Despite the modest accuracy, the results suggest that CNN was able to learn critical emotional cues from low-resolution (48×48) grayscale images. The training and validation curves, shown in Figure 4, demonstrate a steady improvement in both training and validation accuracy over the epochs, with minimal divergence between them. The lower validation loss compared to training loss suggests effective regularization, indicating that the model generalizes well to unseen data. The small gap between validation and training performance confirms limited overfitting. However, fluctuations in training accuracy and loss hint at potential instability, which could be addressed through refined data augmentation strategies and architectural enhancements to improve model robustness and overall generalization [15].

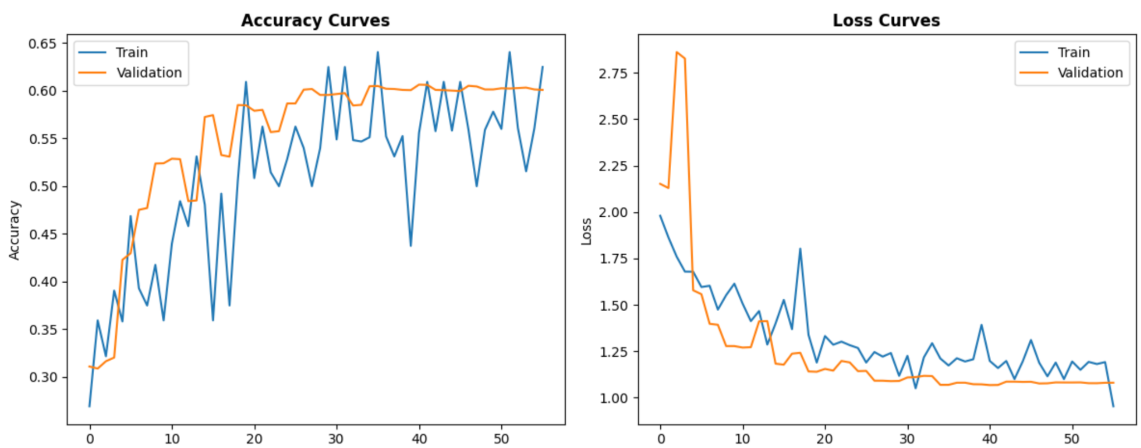

FIGURE 4. Training and validation accuracy and loss curves depicting model learning progress and convergence trends.

Figure 6 presents the accuracy trends for the training, validation, and testing phases. The training accuracy (blue) shows fluctuations but an overall upward trend, while the validation accuracy (green) remains stable and closely follows the training curve, indicating minimal overfitting. The final test accuracy (red dashed line) reaches 60.81%, demonstrating reasonable generalization. However, the instability in training accuracy suggests potential improvements through data augmentation and architectural enhancements. The final test loss of 1.0650 further highlights the model's performance, with scope for refinement.

Real-time deployment was one of the key goals of this study. The implemented system integrates facial emotion recognition with audio feedback using the gTTS (Google Text-to-Speech) API. A screenshot of the real-time application is shown in Figure 5, where the system detects a face, identifies the emotion as "Sleep," and simultaneously outputs the detected state through audible feedback. The use of multithreading ensures that the audio feedback process runs concurrently with frame processing, eliminating delays and ensuring smooth, real-time performance. This feature enhances the system's interactivity and makes it suitable for assistive technologies, such as support systems for visually impaired users or in-vehicle drowsiness detection.

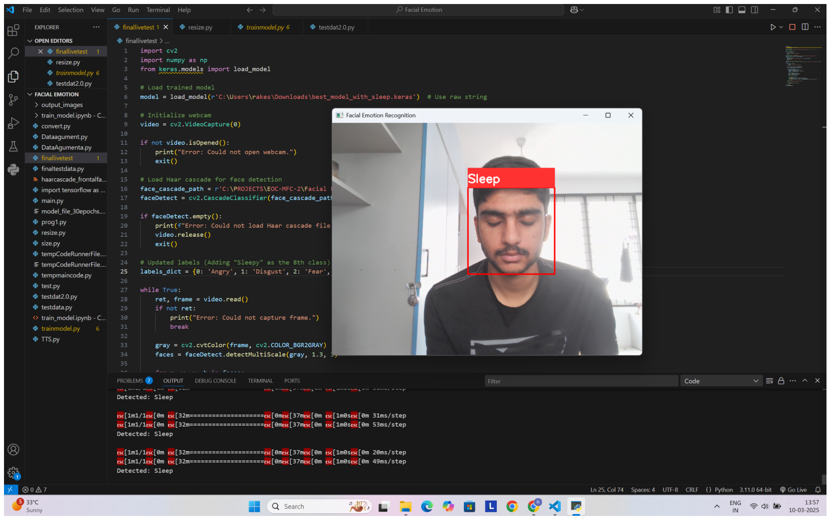

FIGURE 5. Real-time system output showing facial region detection and classification result. In this instance, the model has identified the “Sleep” state, which is overlaid on the video stream and simultaneously converted into speech feedback.

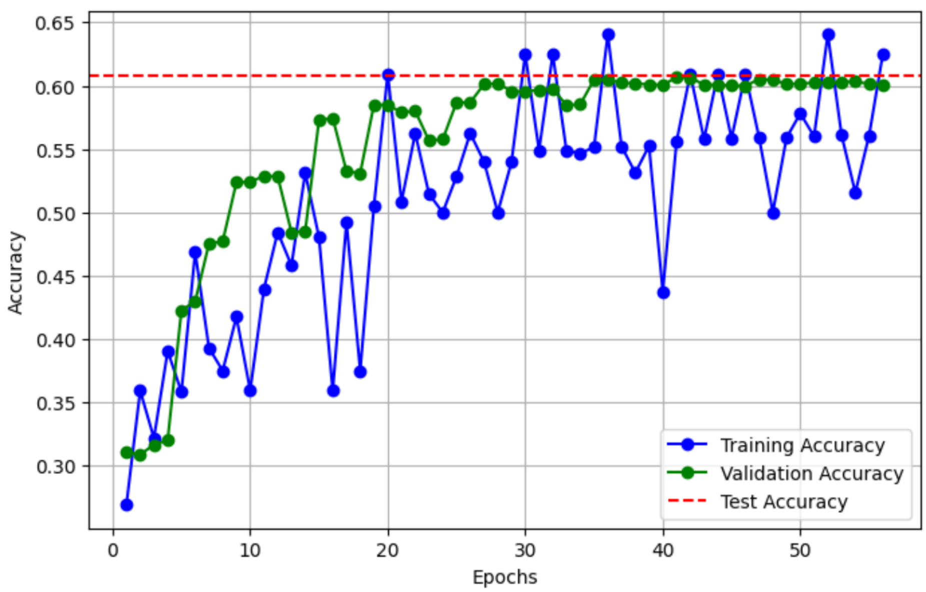

FIGURE 6. Testing accuracy and loss curves depicting model learning progress and convergence trends.

The system completed its designed goals successfully. Firstly, it performed multiclass classification between eight classes, one being the new "Sleep" emotion class. This indicates that the network has the ability to learn expressive and nuanced facial patterns. Secondly, with the introduction of speech-based feedback, the system is now accessible, providing real-time spoken feedback that visually impaired people can use. With the incorporation of the "Sleep" class, the system further grows to cover a greater variety of facial states, which increases the utility of the system in assistive applications.

From a performance standpoint, responsiveness was excellent. The latency between prediction and audio output was negligible due to the asynchronous threading design. The “Sleep” class alone achieved a precision score exceeding 64%, demonstrating practical effectiveness despite being a minority class. Furthermore, the applied augmentation techniques played a pivotal role in improving the recognition of underrepresented categories, particularly for the "Disgust" and "Sleep" classes, which previously suffered from imbalanced representation.

Overall, the system presents a viable solution balancing classification accuracy with real-time usability. Although the accuracy can be further improved with more complex models or additional data, the current implementation strikes a practical balance, making it suitable for deployment in both assistive and safety-critical environments.

V. CONCLUSION

This study successfully developed a real-time Facial Emotion and Sleep Recognition System using a CNN model, achieving a test accuracy of 60.81% with a loss of 1.065, demonstrating reliable generalization across eight emotion classes, including the newly introduced "Sleep" category for drowsiness detection. The model exhibited strong performance for dominant classes like "Happy" (F1-score: 0.81) and "Sleep" (F1-score: 0.91), while struggling with underrepresented emotions such as "Fear" (F1-score: 0.18) due to class imbalance. Techniques like batch normalization, dropout, and data augmentation effectively mitigated overfitting and improved robustness.

Real-time testing confirmed that the system gave instant audio feed to visually impaired individuals and was above 64% accurate in detecting the "Sleep" class, thus proving to be a good classifier for discerning minute facial states. Integration of Haar-based face detection and multithreaded processing guaranteed smooth operation in real-time applications. This study adds value to the area of affective computing with a concern for accessibility. Future work will investigate transfer learning, deeper models, and larger datasets to enhance accuracy, especially for minority classes. Deployment in assistive technology offers a promising avenue for real-world application, crossing accessibility and emotion detection through AI.

REFERENCES

1. K. Jhadi, N. Tiwari, and M. Chawla, "Review of machine and deep learning techniques for expression based facial emotion recognition," in *2024 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS)* (IEEE, 2024) pp. 1–6.
2. A. Kartali, M. Roglić, M. Barjaktarović, M. Đurić Jovičić, and M. M. Janković, "Real-time algorithms for facial emotion recognition: A comparison of different approaches," in *2018 14th Symposium on Neural Networks and Applications (NEUREL)* (2018) pp. 1–4.
3. M. Srivastav, P. Mathur, T. Poongodi, S. Sagar, and S. A. Yadav, "Human emotion detection using open cv," in *2022 2nd International Conference on Innovative Practices in Technology and Management (ICIPTM)*, Vol. 2 (IEEE, 2022) pp. 748–751.
4. S. Rana, R. Chaudhary, M. Gupta, and P. Garg, "Exploring different techniques for emotion detection through face recognition," in *2023 International Conference on Advanced Computing & Communication Technologies (ICACCTech)* (IEEE, 2023) pp. 779–786.
5. A. Nautiyal, D. K. Bhardwaj, R. Narula, and H. Singh, "Real time emotion recognition using image classification," in *Proceedings of the 2023 Fifteenth International Conference on Contemporary Computing* (2023) pp. 8–12.
6. M. K. I. Zim, "Opencv and python for emotion analysis of face expressions," in *2023 3rd International Conference on Innovative Practices in Technology and Management (ICIPTM)* (IEEE, 2023) pp. 1–7.
7. J. Hunt and J. Kim, "Emotion recognition in images and video with python for autism assessment," in *2024 Fifteenth International Conference on Ubiquitous and Future Networks (ICUFN)* (IEEE, 2024) pp. 654–656.
8. C. W. Joseph, G. J. W. Kathrine, S. Vimal, S. Sumathi, D. Pelusi, X. P. B. Valencia, and E. Verdú, "Improved optimizer with deep learning model for emotion detection and classification," Mathematical biosciences and engineering: MBE **21**, 6631–6657 (2024).
9. V. V. Avabratha, S. Rana, S. Narayan, S. Y. Raju, *et al.*, "Speech and facial emotion recognition using convolutional neural network and random forest: A multimodal analysis," in *2024 Asia Pacific Conference on Innovation in Technology (APCIT)* (IEEE, 2024) pp. 1–5.
10. E. Vignesh, S. Srivatsan, and G. Brindha, "Enhancing speech emotion recognition through integrated audio-video analysis," in *2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)* (2025) pp. 1607–1612.
11. S. Sharma, S. Avasthi, I. Malik, and K. Agarwal, "Facial emotion recognition from real-time videos using cnn model," in *2025 2nd International Conference on Computational Intelligence, Communication Technology and Networking (CICTN)* (IEEE, 2025) pp. 50–55.
12. Y. Khaireddin and Z. Chen, "Facial emotion recognition: State of the art performance on fer2013," arXiv preprint arXiv:2105.03588 (2021).
13. S. Depuru, A. Nandam, S. Sivanantham, K. Amala, V. Akshaya, and M. Saktivel, "Convolutional neural network based human emotion recognition system: A deep learning approach," in *2022 Smart Technologies, Communication and Robotics (STCR)* (2022) pp. 1–4.
14. K. Deepa, S. Pradeesh, M. Mathesh, S. Saravanakumar, and R. J. Jefferey, "Adaptive music streaming: Combining cnn facial analysis with snowflake based music categorization," in *2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)* (IEEE, 2025) pp. 1247–1251.
15. B. Kwon, "Data augmentation using convolutional autoencoder for facial emotion recognition," in *2025 International Conference on Electronics, Information, and Communication (ICEIC)* (IEEE, 2025) pp. 1–4.